

AffectFuse: Cross-Task Feature Fusion with Temporal Modeling for Multi-Task Affective Behavior Analysis

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Abstract

Affective behavior recognition in the wild requires joint prediction of continuous valence-arousal, categorical facial expression, and multi-label action units from unconstrained face images. We present our system for the Multi-Task Learning (MTL) track of the 11th Affective Behavior Analysis in-the-wild (ABAW) competition on s-Aff-Wild2, the static selected-frame version of Aff-Wild2. The method focuses on post-encoder adaptation: frozen AffectNet-supervised backbones provide multi-resolution features, while task-specific temporal heads and cross-task fusion modules select the useful signals for each target. For action-unit recognition, we adapt MAE-Face with Low-Rank Adaptation (LoRA) and use DISFA through per-unit expert routing rather than direct sequential transfer. Ablations over backbone, temporal, fusion, and AU-adaptation choices define the final configuration. The final system obtains $\mathcal{P} = 1.7302$ on the official validation split, showing that post-encoder adaptation and task-wise modeling choices provide a strong MTL pipeline without training a new large-scale face foundation model. Code is available [here](#).

1. Introduction

The Affective Behavior Analysis in-the-wild (ABAW) competition series [9, 19, 22–24, 26] studies emotion recognition from faces recorded in unconstrained conditions. Its Multi-Task Learning (MTL) track ties three related problems together on the static selected-frame s-Aff-Wild2 subset of Aff-Wild2 [12, 16, 39]: on roughly 221K frames, one model must regress continuous *valence* and *arousal* [34], classify one of eight *expressions* (the six basic emotions, *Neutral*, and a broad *Other* state), and detect twelve *action units* (AUs) [5]. We denote these tasks as valence–arousal (VA),

expression recognition (EXPR), and AU detection. A single number $\mathcal{P}$ (Eq. (1)) sums the mean valence/arousal concordance correlation coefficient, the macro- F_1 over the eight expressions, and the mean F_1 over the twelve action units. The three tasks reward different structure: expression is a local, discrete identity; valence and arousal drift slowly; and action-unit detection is a sparse multi-label problem dominated by rare units.

Our system focuses on the stages after feature extraction. We use frozen, publicly available AffectNet-supervised face encoders, mainly EfficientNet-B2 and DDAMFN, then build task-specific candidate heads on top of their features. The final configuration is selected through ablation studies over backbone, fusion, temporal, and AU-adaptation choices. Three observations shape this design. First, these frozen supervised features are already strong for expression and valence/arousal, so these tasks benefit more from temporal modeling and cross-task fusion than from retraining the encoder. Second, action units are the task where frozen features stall, so we adapt MAE-Face with LoRA under a collapse-guarded AU objective. Third, because every added module can overfit validation, the ablation path must remain explicit. We make the following contributions.

- **A data-efficient pipeline.** The pipeline built from frozen AffectNet-supervised face encoders, light task heads, and ablation-guided component selection reaches a validation $\mathcal{P}$ of 1.7302 without training a new large-scale face foundation model (Sec. 3).
- **The encoder is not always the bottleneck.** We show that supervised frozen features are strong for expression and valence/arousal, and give a recipe that recovers the after-encoder gains: cross-task feature fusion (Sec. 3.6), a LoRA-adapted action-unit encoder trained under a collapse-guarded loss (Sec. 3.5), and per-unit expert routing that folds in external DISFA data without the domain collapse of sequential fine-tuning (Sec. 3.7).
- **A compact ablation study.** We report the useful validation

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gains and the negative results that mark the edge of what frozen transfer can do on this data (Sec. 4.3).

2. Related Work

ABAW multi-task learning. The challenge has run valence/arousal estimation, expression recognition, and action-unit detection, singly and jointly, across successive editions [9, 10, 14, 17, 19, 22–26], all on the partially labelled s-Aff-Wild2 subset of Aff-Wild2 [12, 13, 15, 16], where many frames carry only a subset of the three label types; here, the prefix “s” denotes the static selected-frame version used for MTL. The organizers also study label-distribution matching for this heterogeneous structure [18, 21] and release in-the-wild analysis toolkits [20]. A 7th-edition MTL system [28] combined a masked-autoencoder extractor, cross-task feature fusion, an LSTM temporal module, and a progressive separate-then-joint training schedule; its ablation shows action-unit features lifting expression and valence/arousal, most on arousal, while fusion slightly hurts the action-unit branch itself. These observations motivate our own fusion ablations. HSEmotion [35, 37] instead used small AffectNet backbones with per-unit adaptive thresholds and inference-time temporal smoothing, the last of which we also adopt.

Frozen backbones, external data, and rare classes. DDAMFN [40] and AffectNet-supervised EfficientNet models [32, 38] give strong in-domain expression and valence/arousal features that we use without further training; a masked autoencoder [6, 41] instead serves here as a fine-tuning target for action units, not a frozen feature source. External corpora are the usual lever for rare action units: BP4D is a common choice in prior work, and we use DISFA [30, 31] for the positives Aff-Wild2 lacks, though we find lab-recorded data transfers poorly when used to initialise the in-the-wild encoder and route it in per column instead. Because rare expressions and sparse AU positives affect the macro- F_1 terms directly, we report class-wise and unit-wise validation diagnostics to expose which labels drive the remaining errors.

3. Method

3.1. Candidate Pipeline

The system overview is shown in Fig. 1. It is a staged collection of task-specific candidate models rather than a single end-to-end multi-task network. Pretrained encoders first produce per-frame features; lightweight heads, temporal models, fusion modules, and ensembles are then evaluated for each task. For validation metric M_t , the selected predictor for task t is

$$f_t^* = \arg \max_{f \in \mathcal{C}_t} M_t(f), \quad \mathcal{P} = \sum_t M_t(f_t^*), \quad (1)$$

Table 1. Pretrained encoders and per-frame outputs.

Encoder	Pretraining	Input	Emb.	EXPR logits
ENET-B2	AffectNet	260 ²	1408	8
DDAMFN	AffectNet	112 ²	512	8
MAE-Face ViT-B	self-sup. faces	224 ²	1536	–

where M_{VA} is mean valence/arousal CCC [27], M_{EXPR} is eight-class macro- F_1 , and M_{AU} is mean F_1 over 12 action units. This task-wise selection is used to organize the ablation study and to choose the final validation configuration.

3.2. Frozen Affective Experts and Features

We use three complementary pretrained face encoders (Tab. 1). The first is `enet_b2_8`, an EfficientNet-B2 from the EmotiEffNet family (ENET) [35, 36, 38]; at 260×260 it returns an embedding $e_i \in \mathbb{R}^{1408}$ and eight expression logits p_i^E . DDAMFN [40] operates at 112×112 and returns $d_i \in \mathbb{R}^{512}$ and logits $p_i^D \in \mathbb{R}^8$. Both models are AffectNet-supervised and remain frozen. Because their output classes use different conventions, both logit vectors are permuted to the Aff-Wild2 challenge order before use.

The main expression representation is

$$x_i^e = [d_i \parallel \tilde{p}_i^D \parallel e_i \parallel \tilde{p}_i^E] \in \mathbb{R}^{1936}, \quad (2)$$

where the tildes denote remapped logits. Initial VA, EXPR, and AU heads use d_i , x_i^e , and e_i , respectively. The third encoder is a self-supervised MAE-Face ViT-B [4, 6, 41]. At 224×224 , its class token and mean patch token are concatenated into a 1536-dimensional feature. MAE-Face is weak as a frozen affect candidate but becomes the AU expert after the adaptation in Sec. 3.5. Frozen outputs are cached once; adapted MAE features are cached after each fine-tuning stage.

3.3. Initial Heads and Objectives

The initial candidates are shallow MLP heads trained on cached features. VA uses a two-output, tanh-bounded regressor on d_i with MSE plus a CCC loss for each axis. EXPR uses an eight-way softmax head on x_i^e , optimized with class-weighted cross-entropy and label smoothing. AU uses 12 independent sigmoid outputs on e_i and a masked asymmetric loss (ASL [1]), a multi-label cross-entropy that down-weights easy negatives to counter the heavy positive/negative imbalance of rare units.

Missing annotations contribute no loss or metric: VA frames marked -5 and EXPR labels marked -1 are ignored, while AU labels marked -1 are masked independently for each column. Thus a frame can supervise any available task or AU without requiring a complete multi-task annotation.

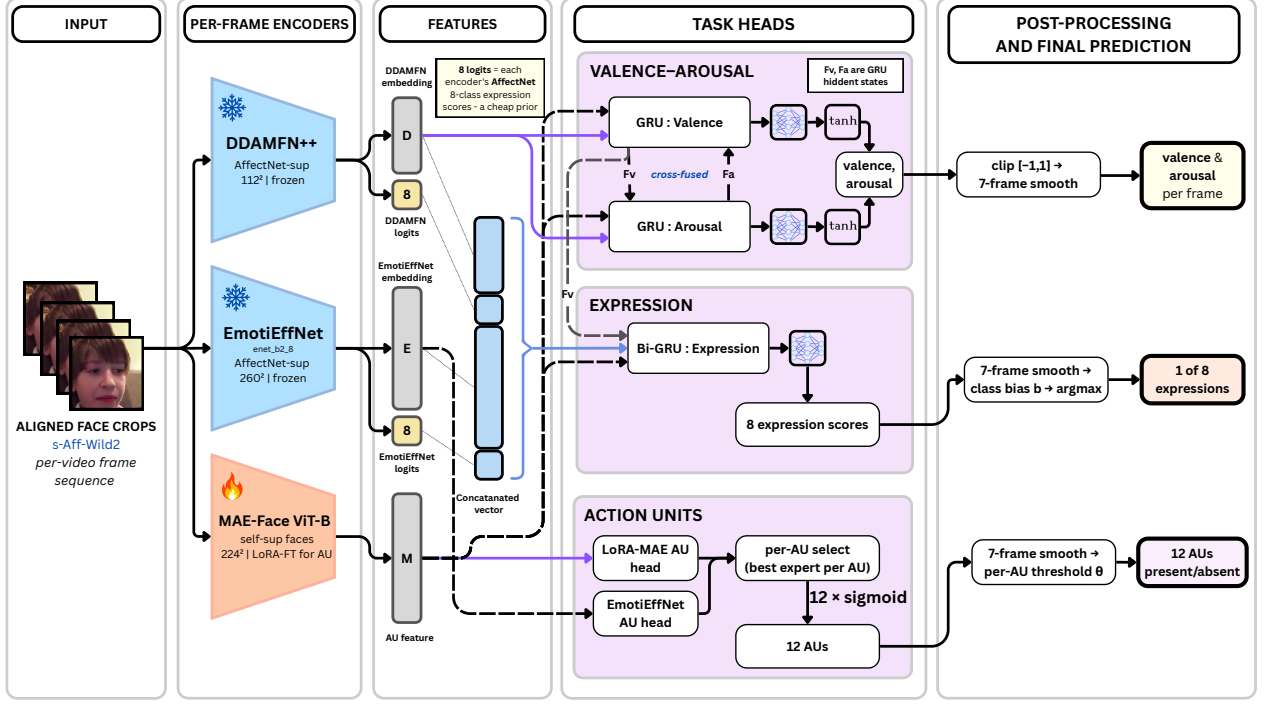


Figure 1. Overview of AffectFuse. Frozen AffectNet-supervised encoders and an AU-adapted MAE-Face encoder produce per-frame features. The AU representation is fused into the temporal VA and EXPR heads, while the AU branch remains independent. A seven-frame moving average and task-specific calibration produce the final predictions.

3.4. Temporal Candidate Heads

For temporal candidates, frames are grouped by video, sorted by frame index, and partitioned into fixed-length windows; the last short window is padded by repeating its final frame. Gated recurrent unit (GRU) [3] and bidirectional GRU heads are evaluated for VA and EXPR. Bidirectional context is valid because inference is offline and the complete recording is available. We also evaluate the Temporal Convergence Module (TCM) described in [28], a two-layer long short-term memory (LSTM) [7] followed by LeakyReLU, on the donor-augmented VA and EXPR features.

Temporal modeling did not provide the same benefit for AU, so retained sequence candidates focus on VA and EXPR. Window length is selected as a task-level hyperparameter: increasing L from 32 to 48 substantially improved the fused temporal candidates, and the final implementation uses $L = 48$ for both tasks.

3.5. Parameter-Efficient AU Adaptation

Frozen features gave limited gains for AU, so we adapt MAE-Face on Aff-Wild2 AU labels. LoRA updates are inserted into the attention QKV/output projections and the two MLP

projections of every transformer block,

$$W' = W + \frac{\alpha}{r} BA, \quad r = 16, \quad \alpha = 32. \quad (3)$$

Here r is the adapter rank, which fixes the size of $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ and hence the parameter budget, while α/r sets the magnitude of the update; we keep LoRA’s two-constant form [8] so that r can be retuned without rescaling the update. All LayerNorm parameters, the AU head, and the complete final four transformer blocks are also optimized. This leaves 31.53M of 89.0M parameters trainable.

Optimization uses layer-wise learning-rate decay of 0.65, linear warmup followed by cosine decay, gradient clipping, image augmentation, and an exponential moving average with decay 0.999. The masked AU loss is

$$\mathcal{L}_{AU} = \mathcal{L}_{ASL} + \lambda(\tau)\mathcal{L}_{sF1}, \quad (4)$$

where $\mathcal{L}_{sF1}$ is a bounded sigmoid- F_1 surrogate [2] with $\beta = 2$, and $\lambda(\tau)$ is ramped in after warmup. An earlier unbounded surrogate collapsed to an all-positive solution. During the ramp, the metric term is therefore disabled for the remainder of training if the validation predicted-positive rate exceeds 0.80 or AU F_1 falls below 0.42.

3.6. Cross-Task Feature Fusion

Let $F_i^{au} \in \mathbb{R}^{1536}$ denote the feature from an AU-adapted MAE-Face encoder. AU serves as a donor to the other tasks. For EXPR, a BiGRU first consumes $[x_i^e \| F_i^{au}]$; a second candidate also receives the pre-classifier valence representation F_i^v :

$$z_i^{e,1} = [x_i^e \| F_i^{au}], \quad z_i^{e,2} = [x_i^e \| F_i^{au} \| F_i^v]. \quad (5)$$

VA is split into separate valence and arousal heads. Both first use $[d_i \| F_i^{au}]$; their pre-classifier states are then cross-fused:

$$z_i^v = [d_i \| F_i^{au} \| F_i^a], \quad z_i^a = [d_i \| F_i^{au} \| F_i^v]. \quad (6)$$

The LSTM-based convergence module is added as another VA/EXPR candidate on the donor-augmented features. We do not feed VA or EXPR features back into the AU branch, since AU fusion reduced validation performance in the corresponding ablations.

3.7. External AU Transfer and Per-Unit Harvest

We prepare DISFA as 75,087 landmark-aligned 224×224 crops from 27 subjects. It annotates 8 of the 12 challenge AUs; AU7, AU10, AU23, and AU24 are stored as -1 and masked. In Stage A, MAE-Face is LoRA-adapted on DISFA using a subject-disjoint holdout. Stage B continues that model on all Aff-Wild2 AUs. The resulting direct AU candidates do not surpass the Aff-Wild2-only expert, consistent with a laboratory-to-in-the-wild domain gap, but they remain in the AU pool and are reused as VA/EXPR donors.

We also train the feature-level candidate ftmix on $[x_i^e \| F_i^{\text{AW}} \| F_i^{\text{D} \rightarrow \text{AW}}]$, where the two MAE features come from the Aff-Wild2-only and DISFA-initialized Stage-B encoders. Although ftmix is not the best complete AU predictor, it remains available to the per-unit harvest. The fusion stack is likewise rerun with the DISFA-initialized donor and with both MAE donors concatenated.

Finally, aumix selects an expert independently for each AU column:

$$r_j^* = \arg \max_{r \in \mathcal{R}} F_{1,j}^{\text{val}}(r), \quad \hat{y}_{i,j}^{\text{AU}} = f_{r_j^*,j}(x_i). \quad (7)$$

Each route retains its expert’s fitted threshold. This design follows the official AU metric, which averages column-wise F_1 , and lets different AU labels use different sources when their validation behavior differs.

3.8. Variance-Reduction Ensembles

The final stage expands the pools with variance-reduced predictors. We rerun the fusion stack with three random seeds and average predictions from matching candidate families. We then form top- K cross-candidate ensembles by averaging the $K = 3$ highest-scoring validation candidates for

each task. Horizontal flip test-time augmentation, defined by averaging original and flipped predictions, is evaluated in the same way.

All variants re-enter the per-task argmax in Eq. (1). The final pool selects a seed ensemble for VA and a top-three ensemble for EXPR, while AU retains aumix . Flip augmentation is not used because it lowers validation scores for all three tasks.

3.9. Post-Processing

Let S_7 denote a centered seven-frame moving average applied separately within each video. The selected outputs are converted to predictions as

$$\begin{aligned} \hat{y}_i^{\text{VA}} &= \text{clip}(S_7(\mathbf{y}^{\text{VA}})_i, -1, 1), \\ \hat{c}_i &= \arg \max_c \{S_7(\ell^{\text{EXPR}})_{i,c} + b_c\}, \\ \hat{a}_{i,j} &= \mathbf{1}[S_7(\sigma(\ell_{i,j}^{\text{AU}})) \geq \theta_j]. \end{aligned} \quad (8)$$

The EXPR bias $b \in \mathbb{R}^8$ and the 12 AU thresholds θ_j are fitted on validation after smoothing. These parameters, together with the selected candidate and AU route map, are saved and reused unchanged at inference.

4. Experiments

4.1. Dataset

For the MTL track, a total of 142,382 training images and 26,876 validation images from s-Aff-Wild2, the static selected-frame version of Aff-Wild2 [11], are provided by the organizers. After removing duplicate entries from the annotation files, the training and validation datasets consist of 141,431 and 26,666 images, respectively. Rows with invalid labels are masked in the corresponding task losses using the official sentinels.

Extra Data. We preprocessed DISFA [31] into 75,087 landmark-aligned crops over 27 subjects; 8 of our 12 AUs are coded there. We additionally acquired and audited **AffectNet+** [33].

4.2. Implementation Details

We used a multi-backbone, multi-resolution feature extraction pipeline for the MTL track. EfficientNet-B2 was evaluated at 260×260 , DDAMFN at 112×112 , and MAE-Face at 224×224 , matching each backbone’s native input setting. The prediction heads were optimized with AdamW using task-appropriate objectives: CCC loss for VA, cross-entropy for EXPR, and asymmetric binary loss for AU detection. All experiments were conducted on an NVIDIA RTX Pro 6000 GPU; complete hyperparameter settings, including batch size, learning rate, scheduler, and optimizer configuration, are provided in the released code.

4.3. Experimental Results

Analysis on the Validation Dataset. We evaluate our final approach to the MTL challenge on the official validation dataset. Table 2 gives the main validation score. Figure 2 shows the EXPR confusion matrix on the same split.

Table 2. Quantitative results on the officially provided validation dataset.

Method	Avg. CCC	F_{expr}	F_{AUs}	P
Organizer score	—	—	—	0.4500
Ours	0.6596	0.5097	0.5609	1.7302

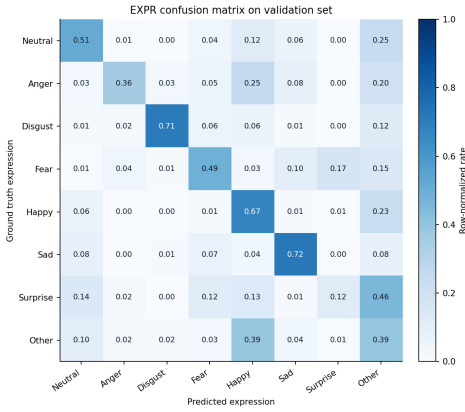


Figure 2. EXPR confusion matrix on validation set.

Table 3 summarizes the complete validation path. Starting from frozen backbone features and plain task heads, the ablation adds one family of candidates at each stage and reports the selected task-wise configuration. The largest gains come from temporal modeling, cross-task feature fusion, and the final variance reduction step, while AU-specific changes mainly stabilize the action-unit branch.

Table 3. Validation progression.

Stage	VA	EXPR	AU	P	Δ
Frozen backbones, plain heads	0.5294	0.3948	0.5287	1.4529	—
AU on concat. features + ASL	0.5870	0.4165	0.5401	1.5436	+0.091
AU: LoRA fine-tuned MAE	0.5870	0.4165	0.5573	1.5608	+0.017
GRU tuning	0.6188	0.4314	0.5573	1.5969	+0.036
Decoupled window + BiGRU	0.6235	0.4477	0.5573	1.6284	+0.032
Cross-task feature fusion	0.6260	0.4841	0.5573	1.6674	+0.039
DISFA + re-fusion	0.6369	0.4841	0.5573	1.6783	+0.011
Per-AU mixture of experts	0.6361	0.4838	0.5659	1.6858	+0.008
Temporal window 32 $\rightarrow$ 48	0.6526	0.5035	0.5609	1.7170	+0.031
Multi-seed + top-3 ensembling	0.6596	0.5097	0.5609	1.7302	+0.013

The following ablations unpack the main rows of Table 3: backbone choice, MAE-Face adaptation, fusion, external AU data, temporal window length, and ensembling.

Impact of Backbone Diversity. Table 4 compares the frozen feature sets used before temporal modeling. The two AffectNet backbones gave the best pair; extra frozen backbones did not improve validation scores.

Table 4. Backbone comparison.

Candidate	EXPR	AU	Decision
EfficientNet-B2+DDAMFN++	0.4165	0.5401	used
+ POSTER++ [29]	0.4164	0.5373	no gain
MAE-Face, frozen	0.3770	—	low EXPR

POSTER++ changed EXPR by only -0.0001 and reduced AU, so it was not used in the final frozen feature set. Frozen MAE-Face was also weaker for EXPR before task-specific adaptation.

Impact of Fine-tuning the MAE Feature Extractor. Table 5 shows the AU result after LoRA fine-tuning MAE-Face.

Table 5. LoRA fine-tuning MAE-Face for AU.

Unfreeze	Params	AU	P	Gain
2	15.9 M	0.5485	1.5520	+0.0084
4	30.1 M	0.5519	1.5554	+0.0118
6	~ 37 M	0.5573	1.5608	+0.0172
8	~ 44 M	0.5544	1.5579	+0.0143

Six unfrozen blocks gave the best AU score. The same MAE fine-tuning did not help EXPR or VA (Table 6); the EXPR run saturated at 0.4012, below our frozen feature ensemble. Figure 3 shows the AU stats at the unit level on the validation set.

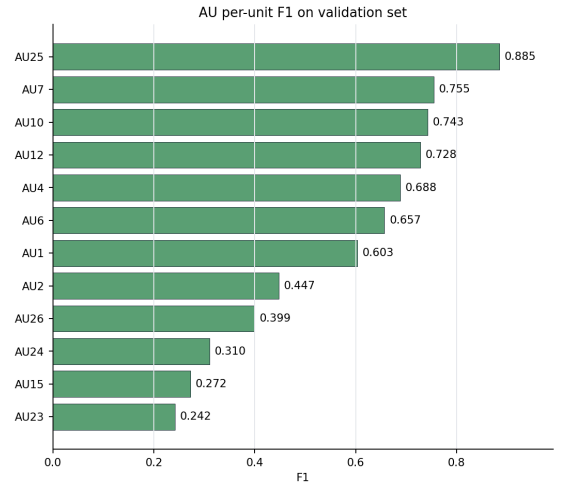


Figure 3. AU per-unit F1 on validation set.

Table 6. Fine-tuning MAE-Face for EXPR and VA.

Task	ft-MAE	+ concat	+ temp.	Frozen best
EXPR	0.3667	0.3909	0.4012	0.4477
VA	0.4332	0.4429	0.5579	0.6235

Impact of Feature Fusion. Table 7 shows the useful candidates introduced at the fusion stage.

Table 7. Fusion-stage candidates.

Task	Before	Best candidate	Gain
EXPR	0.4477	0.4841 ($F_{au} + F_v$)	+0.0364
VA	0.6235	0.6260 (TCM)	+0.0025

Fusion mainly helped EXPR. AU features alone were not enough, but combining AU and valence features raised EXPR from 0.4477 to 0.4841. VA improved slightly through the temporal convergence module rather than cross-task donors. AU was kept unfused because all fusion variants were below 0.5573.

Impact of External AU Data. Table 8 evaluates sequential DISFA-to-Aff-Wild2 AU transfer, where the final comparison is measured on the official Aff-Wild2 validation split.

Table 8. Impact of DISFA sequential transfer.

Training path	AU
Stage-A, held-out DISFA subjects	0.7280
Stage-B on Aff-Wild2 (lr $\times 0.4$)	0.5451
Aff-Wild2-only training	0.5573

DISFA pre-training worked on held-out DISFA subjects, but after transfer to Aff-Wild2 it did not surpass the Aff-Wild2-only AU model. We therefore did not use this branch as the final AU classifier; its useful role was as an additional donor in the later re-fusion step.

Impact of the Temporal Module. Table 9 shows the effect of window size and hidden dimension before fusion.

Table 9. Temporal module before fusion.

W / H	VA	EXPR
32 / 256	0.6016	0.4149
32 / 512	0.6188	0.4314
32 / 768	0.6086	0.4295
48 / 512	0.6235	0.4105

Hidden size 512 was best in this sweep. A longer window helped VA but reduced raw EXPR, so later heads were allowed to use task-specific windows.

Table 10. Temporal window after fusion.

Task	$W=32$	$W=48$	Δ
EXPR	0.4838	0.5035	+0.020
VA	0.6361	0.6526	+0.017
AU	0.5659	0.5609	-0.005
P	1.6858	1.7170	+0.031

On the fused pipeline, increasing the window to 48 improved VA and EXPR and raised P by +0.031. This shows that the best temporal length depends on the feature representation, not only on the task.

Impact of Variance Reduction. Table 11 reports the final ensembling step.

Table 11. Multi-seed, top- K , and flip-TTA.

Task	Before	Seed/top- K	Flip-TTA	Selected
VA	0.6526	0.6596	0.6577	3-seed ens.
EXPR	0.5034	0.5097	0.5040	top-3 ens.
AU	0.5609	0.5609	0.5606	unchanged
P	1.7168	1.7302	-	

Multi-seed and top- K ensembling improved VA and EXPR. Flip-TTA was lower for all tasks, likely because expressions and AUs can be asymmetric after a horizontal flip. We therefore excluded flip-TTA and retained the multi-seed/top- K configuration, which gave the best validation score of $P = 1.7302$.

5. Conclusion

A data-efficient pipeline of frozen AffectNet-supervised face encoders, cross-task feature fusion, a LoRA-adapted action-unit expert, and per-unit routing of external data reaches a validation $\mathcal{P}$ of 1.7302 on the 11th ABAW MTL track, with strong valence/arousal and expression performance, without training a new large-scale face foundation model or relying on a licensed AU corpus. The system is built through a staged ablation framework that compares backbone, fusion, temporal, AU-adaptation, and ensembling choices under the same validation protocol. Two findings carry beyond our entry. The encoder is not always the bottleneck for valence/arousal and expression, since a fine-tuned autoencoder saturates below the frozen supervised ensemble; effort is better spent after the encoder, on action units. The remaining validation errors point to action-unit coverage and expression-class

confusion as the main residual issues. Strong frozen features combined thoughtfully after the encoder, and selected through explicit ablations, offer a reproducible route without training a new large-scale face foundation model.

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